Nathan Glymph

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Education

North Carolina State University **August 2025 - May 2027** (Expected)
M.S., Electrical Engineering;

- **Concentrations:** control systems, signals & imaging, and bioelectronics

North Carolina State University & UNC Chapel Hill **August 2022 - May 2026**
Biomedical Engineering; Minor: Chinese Studies 3.9 GPA

Work and Volunteer Experience

Process Engineering Intern, Plexus Corporation **May 2026 – August 2026**

- Performed PCBA and high-level assembly for surgical robotics and other medical devices. Investigated SMT defects and created process improvements to improve output efficiency, ensure operator safety, and uphold quality under regulatory standards such as IPC-A-600/610, ISO 9001, and ISO 13485
- Created a web app via Google Appscript to process solder paste inspector (SPI) measurements and end of line defects to calculate optimal inspection settings for the SPI, reducing defect escapes by 87% and false calls by 72%.
- Designed design assembly fixtures to reduce product variation, improve output efficiency, and mitigate safety risks.

Biomedical Equipment Technician, EWB Summer Institute **June 2024 – August 2024**

- Diagnosed and repaired broken medical equipment in low resource hospitals in Guatemala including pulse oximeters, patient monitors, infant warmers, suction pumps, infusion pumps, and patient beds.
- Designed and built a system of low-cost portable privacy curtains for the intensive care unit at Hospital Nacional de Mazatenango; designed a hospital map for visitors to improve traffic flow and reduce congestion

Engineering Intern, ACT Power Services **May 2023 – May 2024**

- Performed operations and management for large-scale solar projects alongside team of engineers
- Created monthly and annual system performance report deliverables for clients by performing work order investigations
- Identified and reported on underperforming hardware such as inverters, combiner boxes, and trackers by analyzing long term production trends using remote site-monitoring software

Skills

- **CAD/CAM Software:** Fusion 360 (PCB), KiCAD, SolidWorks (assemblies, drawings), Onshape, GerbTool
- **Design for Machining:** CNC Milling, CNC Lathing, acrylic engraving, 3D printing (FDM/SLS)
- **Embedded Systems:** STM32 (STM32CubeIDE/MX), Atmel Microcontrollers (Microchip Studio – C)
- **Programming:** MATLAB / Simulink, Python, C
- **Microsoft/Google Workspace:** spreadsheet/report automation via Appscript, Apps Sheet, Data Studio, Python
- **Languages:** Spanish (proficient), Chinese (basic conversational)

Honors and Activities

Chief Technology Officer, Helping Hand Project at NCSU **August 2022 – May 2026**

- Co-led chapter of non-profit organization that designs, builds, and donates prosthetic arms for children with limb differences using CAD (Onshape, Solidworks) and 3D printing
- Led team which performs maintenance and repairs on 3D printers and optimizes all club projects for 3D printing
- Led a group in designing a finger prosthesis as a case team leader

Eagle Scout, Boy Scouts of America **August 2015 – May 2022**

- Designed a compost system for a community garden and led a team of peers in its construction and installation